

from which

$$pK_b = -\log[K_b] \quad (4.5)$$

The stronger the base, the weaker is its conjugate acid.

Because water is amphiprotic, a proton transfer equilibrium exists even in the absence of added acids or bases. The proton transfer from one water molecule to another is called autoprotolysis (or 'autoionization'). The extent of autoprotolysis and the composition of the solution at equilibrium is described by the autoprotolysis constant (or 'autoionization constant') of water:

$$K_w = [H_3O^+][OH^-] = 10^{-14} \frac{\text{mol}^2}{\text{dm}^6} \quad (4.6)$$

Classification

A substance is classified as a strong acid if the proton transfer equilibrium lies strongly in favour of donation of a proton to the solvent. Thus, a substance with $pK_a < 0$ (corresponding to $K_a > 1$ and usually to $K_a \gg 1$) is a strong acid.

A substance with $pK_a > 0$ (corresponding to $K_a < 1$) is classified as a weak acid.

Check out Figure 4.4 for the effective pH in water for different materials. There are three classes of acids to consider:

- An aqua acid, in which the acidic proton is on a water molecule coordinated to a central metal ion.

$$E(OH_2)(aq) + H_2O(l) \rightleftharpoons E(OH)^-(aq) + H_3O^+(aq)$$
- A hydroxoacid, in which the acidic proton is on a hydroxyl group without a neighbouring oxo group (=O). An example is $Te(OH)_6$
- An oxoacid, in which the acidic proton is on a hydroxyl group with an oxo group attached to the same atom. Sulfuric acid, H_2SO_4 ($O_2S(OH)_2$), is an example of an oxoacid.

Check out plots 3,4,5 on page 122.

Check out Table 4.3.

Pauling's rules

The trends can be systematized semiquantitatively by using two empirical rules devised by Linus Pauling, where p is the number of oxo groups and q is the number of hydroxyl groups:

- For the oxoacid $O_pE(OH)_q$,

$$pK_a \approx 8 - 5p$$

- The successive pK_a values of polyprotic acids (those with $q > 1$), increase by 5 units for each successive proton transfer.

Check out section "4.8 Nonaqueous solvents".

4.1.2 Lewis acidity

- A Lewis acid is an electron pair acceptor.
- A Lewis base is an electron pair donor.

Check out "4.9 Examples of Lewis acids and bases" for examples.

Hard and soft acids and bases are identified empirically by the trends in stabilities of the complexes that they form: hard acids tend to bind to hard bases and soft acids tend to bind to soft bases.

- Hard acids bond in the order: $I < Br < Cl < F$.
- Soft acids bond in the order: $F < Cl < Br < I$.

4.2 Applications of acid-base chemistry

A superacid is a substance that is a more efficient proton donor than pure sulfuric acid.

They are formed when a powerful Lewis acid is dissolved in a powerful Brønsted acid.

The most common superacids are formed when SbF_5 is dissolved in fluorosulfonic acid, HSO_3F , or anhydrous HF .

Superacids are known that can protonate almost any organic compound. In the 1960s, George Olah and his colleagues found that carbonium ions were stabilized when hydrocarbons were dissolved in superacids.

A superbases is a compound that is a more efficient proton acceptor than the OH^- ion, the strongest base that can exist in aqueous solution.

4.3 Exercises

Exercise 4.1 — 1 point. Identify the conjugate bases corresponding to the following acids:

- (a) $[Co(NH_3)_5(OH_2)]^{3+}$
- (b) HSO_4^-
- (c) CH_3-OH
- (d) $H_2PO_4^-$
- (e) $Si(OH)_4$
- (f) HS^-

Exercise 4.2 — 1 point. Arrange the oxides (a) Al_2O_3 , (b) B_2O_3 , (c) BaO , (d) CO_2 , (e) Cl_2O_7 , (f) SO_3 in order from the most acidic through amphoteric to the most basic.

Exercise 4.3 — 1 point. Calculate the equilibrium concentration of H_3O^+ in a 0.10 M solution of butanoic acid ($K = 1.86 \cdot 10^{-5}$). What is the pH of this solution?

Exercise 4.4 — 1 point. Use Pauling's rules to place the following acids in order of increasing acid strength HNO_2 , H_2SO_4 , $HBrO_3$, and $HClO_4$ in a nonlevelling solvent.

Exercise 4.5 — 1 point. Explain why hydrogen selenide is a stronger acid than hydrogen sulfide.